

THE EVEN-ORDER GRÜNSCHLOSS–KELLER PERMUTATION NETS ARE $(0, m, 2)$ -NETS

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ABSTRACT. Grünschloß and Keller (2009) constructed two-dimensional base-2 point sets with large minimum toroidal distance by a nonlinear permutation rule. For odd m they proved that the 2^m points form a $(0, m, 2)$ -net. For even m they verified the net property by computer through $m = 22$ and left a proof open. We prove that the even-order construction is a $(0, m, 2)$ -net for every even $m \geq 4$. The proof writes the first coordinate as an exact concatenation of $\lfloor k/4 \rfloor$ and a bit-reversal term and inverts every dyadic prefix map explicitly. For the row-shifted tiling metric of the source we prove that the minimum squared distance on the integer scale (coordinates in $[0, 2^m)$) equals $241 \cdot 2^{m-8}$ for every even $m \geq 8$; this is the value the source reported from computation for $8 \leq m \leq 22$. Equality holds for exactly 2^{m-2} pairs, namely the pairs $\{(k, j), (k', j)\}$ of diagonal indices with $\text{rev}_r(k') = \text{rev}_r(k) + 1$ and $\text{rev}_r(k) \equiv 1 \pmod{4}$. A Lean 4 development checks the net theorem for every even $m \geq 4$, the upper bound for every even $m \geq 8$, and the exact minimum at $m = 8$ over all pairs and lattice translates, using only the standard axioms. Three independent exact generators reproduce the point sets and the distances, with all pairs through $m = 16$ and the reduced cases of the proof through $m = 30$.

1. INTRODUCTION

A $(0, m, 2)$ -net in base 2 is a set of 2^m points in $[0, 1)^2$ such that every dyadic elementary interval of area 2^{-m} contains exactly one point [5, 12]. Grünschloß, Hanika, Schwede, and Keller [7] and Grünschloß and Keller [8] searched for $(0, m, 2)$ -nets whose minimum toroidal distance is as large as possible, motivated by sampling in image synthesis, and gave a construction that is a permutation of a diagonal pattern rather than a digital net. The second paper proves the net property for odd m (its Theorem 1) and gives a modified construction for even m , whose net property was verified by computer for even $m \leq 22$. Section 3.2.2 of [8] reads: “While we believe that a similar proof that $t = 0$ is possible as for the odd m case, we only verified the $t = 0$ property using a computer program for all even $m \leq 22$.

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The minimum toroidal distance equals $\sqrt{241}$ for $m = 8$ and $\sqrt{241 \cdot 2^{m-8}}/2^m$ for all even m where $10 \leq m \leq 22$. [...] We would like to stress that the modified minimum toroidal distance measure that respects the shifts of the rows in the tiling has been used for these measurements.”

We prove both statements. Theorem 3.2 shows that the even-order construction is a $(0, m, 2)$ -net for every even $m \geq 4$. Theorem 4.1 shows that the minimum squared distance under the row-shifted tiling metric is $241 \cdot 2^{m-8}$ on the integer scale for every even $m \geq 8$, and identifies the 2^{m-2} pairs that attain it. Once the first coordinate is read as the concatenation $\lfloor k/4 \rfloor + Sz$ of Lemma 3.1, the net proof follows the pattern of the source’s odd-order proof, with the boundary width $a = r + 1$ treated separately; the distance theorem and its equality set have no counterpart in the source.

Section 5 describes the Lean 4 formalization (Theorem 3.2 for every even $m \geq 4$, the upper bound of Theorem 4.1 for every even $m \geq 8$, and the exact minimum at $m = 8$) and the three checkers. The dyadic-net literature after [8] has classified and optimized matrix-generated nets [1, 2] and has studied the separation and quasi-uniformity of digital nets [4, 13, 14]; the construction treated here is not a digital net, nor a digital shift of one (Section 2). The survey of Keller, Wächter, and Binder [9] cites the construction without a proof of the even case. After exact-phrase, citation, author, repository, preprint, and archive searches completed on 3 September 2026, we found no prior proof of the even-order net property or of the distance formula. This is a dated search record, not a claim of absolute priority.

2. THE CONSTRUCTION AND THE TILING METRIC

Throughout, $m = 2r$ is even with $r \geq 2$ (the header’s even branch shifts by $r - 2$ bits, so $m = 2$ is not generated by it), and

$$Q = 2^r, \quad S = Q/4 = 2^{r-2}, \quad N = 2^m = Q^2.$$

For $0 \leq k < Q$, $\text{rev}_r(k)$ denotes the integer obtained by reversing the r binary digits of k ; in particular $\text{rev}_r(2^i) = 2^{r-1-i}$ for $0 \leq i < r$. The map rev_r is an involution of $\{0, \dots, Q - 1\}$.

2.1. The point set. The published header `gendiag0m2.h` [6] generates, for even m , the N integer points indexed by diagonals k and positions j , $0 \leq k, j < Q$:

- (1) $x_{k,j} = (Q \text{rev}_r(k) + \lfloor k/4 \rfloor + S(j + 1)) \bmod N,$
- (2) $y_{k,j} = k + Qj.$

The header computes the first term as the 32-bit van der Corput bit reversal of k shifted right by $32 - m$; for $k < Q$ this equals $Q \text{rev}_r(k)$, because the top r bits of k as an m -bit word are zero. In the words of [8], the point with smallest second coordinate on diagonal d is $(2^m \phi_2(d) + \lfloor d/4 \rfloor + 2^{m/2-2}, d)$,

and the points on a diagonal are spaced by $(2^{m/2-2}, 2^{m/2})$; this is (1)–(2) with $d = k$ and $2^m \phi_2(k) = Q \operatorname{rev}_r(k)$. The normalized point set is

$$P_m = \{(x_{k,j}/N, y_{k,j}/N) : 0 \leq k, j < Q\} \subset [0, 1]^2.$$

The point $(0, 0)$ is not in P_m , so P_m is not a digital net in the sense of [5]; it is not a digital shift of one either, since the $\mathbb{F}_2$ -span of the differences $w(p) \oplus w(p_0)$ of the $2m$ -bit coordinate words $w(p) = 2^m x + y$ has dimension $3m/2$ for even $4 \leq m \leq 16$ (computed), whereas a digitally shifted digital net has difference span of dimension m . Since $(k, j) \mapsto y_{k,j}$ is a bijection onto $\{0, \dots, N-1\}$, the N points are distinct.

2.2. Elementary intervals. For $0 \leq a \leq m$ the dyadic elementary intervals with a prescribed leading bits in the first coordinate and $m-a$ in the second are indexed by pairs (b_x, b_y) with $0 \leq b_x < 2^a$ and $0 \leq b_y < 2^{m-a}$. An integer point (x, y) with $0 \leq x, y < N$ lies in the interval $[b_x 2^{-a}, (b_x + 1) 2^{-a}) \times [b_y 2^{a-m}, (b_y + 1) 2^{a-m})$ after division by N exactly when

$$(3) \quad \lfloor x/2^{m-a} \rfloor = b_x \quad \text{and} \quad \lfloor y/2^a \rfloor = b_y.$$

Define $F_a(k, j) = (\lfloor x_{k,j}/2^{m-a} \rfloor, \lfloor y_{k,j}/2^a \rfloor)$. Both the index set and the set of intervals have N elements, so P_m is a $(0, m, 2)$ -net in base 2 exactly when every F_a is injective.

2.3. The row-shifted tiling metric. Tile (p_x, p_y) of the header's routine `getShiftedTiling` contains the points $((x + p_y N/4) \bmod N + p_x N, y + p_y N)$; each tile row is shifted by $N/4 = 2^{m-2}$ relative to the row below, as in Figure 6 of [8]. The union of all tiles is $P_m^{\text{int}} + \Lambda_m$ for the translation lattice

$$\Lambda_m = \langle (N, 0), (N/4, N) \rangle_{\mathbb{Z}},$$

where P_m^{int} is the integer point set. The modified minimum toroidal distance of the source is the minimum distance between distinct points of this union. For integer points $p = (x, y)$ and $p' = (x', y')$ put

$$(4) \quad d_*^2(p, p') = \min_{A, B \in \mathbb{Z}} [(x' - x + AN + BN/4)^2 + (y' - y + BN)^2].$$

Translates of one point by nonzero lattice vectors are at distance at least N , so the minimum over the tiling equals $\min_{p \neq p'} d_*(p, p')$. The ordinary toroidal metric uses the lattice $\langle (N, 0), (0, N) \rangle$ instead; it enters only in Section 6.

2.4. Reading the source. The source states the distance for $m \geq 10$ with the radical covering $241 \cdot 2^{m-8}$ (verified on the rendered page; a plain-text extraction loses the extent of the radical), so on the integer scale it reads $d_*^2 = 241 \cdot 2^{m-8}$. Table 1 of [8] lists the squared integer-scale minima 13, 53, 241, 964, 3856, 15424, 61696 for $m = 4, 6, \dots, 16$; the values for $m \geq 8$ are $241 \cdot 2^{m-8}$, and the values 13 and 53 at $m = 4, 6$ are not. The opposite row shift, with lattice $\langle (N, 0), (-N/4, N) \rangle$, gives 10 instead of 13 at $m = 4$

and the same values for $6 \leq m \leq 16$, so Table 1 of the source also pins the orientation of the tiling.

3. THE NET PROPERTY

Lemma 3.1 (Concatenation). *Let $q_k = \lfloor k/4 \rfloor$ and let $z_{k,j} \in \{0, \dots, 4Q-1\}$ be the residue of $j+1+4\text{rev}_r(k)$ modulo $4Q$. Then $x_{k,j} = q_k + Sz_{k,j}$.*

Proof. Since $Q = 4S$, the sum in (1) equals $q_k + S(j+1+4\text{rev}_r(k))$, and $S \cdot 4Q = N$. Reducing modulo N replaces $j+1+4\text{rev}_r(k)$ by $z_{k,j}$, and $0 \leq q_k < S$, $0 \leq z_{k,j} < 4Q$ give $0 \leq q_k + Sz_{k,j} < N$, so no further reduction occurs. $\square$

Thus the low $r-2$ bits of $x_{k,j}$ are those of q_k and the high $r+2$ bits are those of $z_{k,j}$. The second fact needed is the behavior of reversal on concatenations: for $0 \leq \tau < 2^a$ and $0 \leq K < 2^p$,

$$(5) \quad \text{rev}_{a+p}(2^a K + \tau) = 2^p \text{rev}_a(\tau) + \text{rev}_p(K).$$

Theorem 3.2. *For every even $m \geq 4$, P_m is a $(0, m, 2)$ -net in base 2.*

Proof. Fix a with $0 \leq a \leq m$. We show that $F_a(k, j) = F_a(k', j')$ forces $(k, j) = (k', j')$.

Case $a \leq r$. From (2), $\lfloor y_{k,j}/2^a \rfloor = 2^{r-a}j + \lfloor k/2^a \rfloor$ with $\lfloor k/2^a \rfloor < 2^{r-a}$, so equal second components give $j = j'$ and $K := \lfloor k/2^a \rfloor = \lfloor k'/2^a \rfloor$. Write $k = 2^a K + \tau$, $k' = 2^a K + \tau'$ with $0 \leq \tau, \tau' < 2^a$. By Lemma 3.1 and $2^{m-a} = S \cdot 2^{r+2-a}$, the first component of F_a is $\lfloor z_{k,j}/L \rfloor$ with $L = 2^{r+2-a}$. By (5) with $p = r-a$,

$$z_{k,j} \equiv C + L \text{rev}_a(\tau) \pmod{4Q}, \quad C = j+1+4\text{rev}_{r-a}(K),$$

and $4Q = L2^a$. Writing $C = C_0 + LC_1$ with $0 \leq C_0 < L$ gives $\lfloor z_{k,j}/L \rfloor = (C_1 + \text{rev}_a(\tau)) \bmod 2^a$. The same holds for τ' with the same C_1 . Equal first components therefore give $\text{rev}_a(\tau) = \text{rev}_a(\tau')$, hence $\tau = \tau'$ and $k = k'$.

Case $a = r+1$. Now $\lfloor y_{k,j}/2^{r+1} \rfloor = \lfloor j/2 \rfloor =: J$, so J is common to both points; write $j = 2J + u$, $j' = 2J + u'$ with $u, u' \in \{0, 1\}$. By Lemma 3.1, $\lfloor x_{k,j}/2^{r-1} \rfloor = \lfloor z_{k,j}/2 \rfloor$, and

$$\lfloor z_{k,j}/2 \rfloor = (J + 2\text{rev}_r(k) + u) \bmod 2Q.$$

Since $2\text{rev}_r(k) + u < 2Q$, equal first components give $2\text{rev}_r(k) + u = 2\text{rev}_r(k') + u'$. Parity gives $u = u'$, then $\text{rev}_r(k) = \text{rev}_r(k')$, so $k = k'$ and $j = j'$.

Case $a = r+c$ with $2 \leq c \leq r$. Now $\lfloor y_{k,j}/2^a \rfloor = \lfloor j/2^c \rfloor =: J$ is common; write $j = 2^c J + u$, $j' = 2^c J + u'$ with $0 \leq u, u' < 2^c$. Put $a_0 = r-c+2$ and write $k = 2^{a_0}h + \ell$ with $0 \leq \ell < 2^{a_0}$, so $h = \lfloor k/2^{a_0} \rfloor < 2^{c-2}$. Dividing the concatenation of Lemma 3.1 by 2^{r-c} gives

$$\lfloor x_{k,j}/2^{r-c} \rfloor = h + 2^{c-2}z_{k,j},$$

an exact concatenation because $h < 2^{c-2}$. Equal first components therefore give $h = h'$ and $z_{k,j} = z_{k',j'}$. By (5), $\text{rev}_r(k) = 2^{c-2} \text{rev}_{a_0}(\ell) + \text{rev}_{c-2}(h)$, so

$$z_{k,j} \equiv 2^c(J + \text{rev}_{a_0}(\ell)) + w \pmod{2^{r+2}}, \quad w = u + 1 + 4 \text{rev}_{c-2}(h).$$

Reducing modulo 2^c (which divides 2^{r+2}) gives $z_{k,j} \equiv w \pmod{2^c}$; since $h = h'$, the equality $z_{k,j} = z_{k',j'}$ yields $u \equiv u' \pmod{2^c}$, hence $u = u'$ and $w = w'$. Dividing by 2^c then gives

$$\lfloor z_{k,j}/2^c \rfloor = (J + \text{rev}_{a_0}(\ell) + \lfloor w/2^c \rfloor) \pmod{2^{a_0}},$$

so $\text{rev}_{a_0}(\ell) = \text{rev}_{a_0}(\ell')$, hence $\ell = \ell'$, $k = k'$, and $j = j'$.

In every case F_a is injective, hence bijective, and P_m is a $(0, m, 2)$ -net. $\square$

4. THE TILING DISTANCE

Theorem 4.1. *For every even $m \geq 8$,*

$$\min_{p \neq p' \in P_m^{\text{int}}} d_*^2(p, p') = 241 \cdot 2^{m-8}.$$

On the integer scale the minimum distance is $\sqrt{241} \cdot 2^{(m-8)/2}$, and after normalization by 2^m it is $\sqrt{241 \cdot 2^{m-8}}/2^m$. The minimum is attained exactly by the 2^{m-2} unordered pairs $\{(k, j), (k', j')\}$ with $\text{rev}_r(k') = \text{rev}_r(k) + 1$ and $\text{rev}_r(k) \equiv 1 \pmod{4}$, each realized with $B = 0$ in (4), that is, within one tile row, and with $A \neq 0$ exactly when the first coordinates wrap modulo N .

Proof. Put $T = Q/16$, so $N = 256T^2$ and $241 \cdot 2^{m-8} = 241T^2$.

Upper bound. Take $(k, j) = (Q/2, 0)$ and $(k', j') = (Q/4, 0)$. Since $\text{rev}_r(Q/2) = 1$ and $\text{rev}_r(Q/4) = 2$, equation (1) gives $x_{k,j} = 22T$ and $x_{k',j'} = 37T$, while $y_{k,j} = 8T$ and $y_{k',j'} = 4T$. The difference $(15T, -4T)$ has squared length $225T^2 + 16T^2 = 241T^2$.

Reduction of the lower bound. Consider distinct points indexed by (k, j) and (k', j') , and write

$$\delta = k' - k, \quad s = j' - j, \quad \Delta q = \lfloor k'/4 \rfloor - \lfloor k/4 \rfloor, \quad \Delta R = \text{rev}_r(k') - \text{rev}_r(k).$$

For a lattice translate with coefficients (A, B) in (4) set $e = s + BQ$. Since $N/4 = SQ$, the translated difference (X, Y) satisfies

$$(6) \quad Y = \delta + Qe, \quad X \equiv Q\Delta R + \Delta q + Se \pmod{N},$$

where, for fixed B , minimizing over A makes X the residue of least absolute value, and under $|X| < Q \leq N/2$ it is the unique such residue. Suppose $X^2 + Y^2 \leq 241T^2$. As $241T^2 < Q^2$, both $|X| < Q$ and $|Y| < Q$. From $|\delta| \leq Q - 1$ we get $|e| \leq 1$. Let ρ be the least absolute residue of ΔR modulo Q , so that $Q\Delta R \equiv Q\rho \pmod{N}$ and $|Q\rho| \leq N/2$. Since $|\Delta q| \leq Q/4 - 1$ and $|Se| \leq Q/4$, the distance from X to $N\mathbb{Z}$ is at least $Q|\rho|$ minus $Q/2 - 1$:

$$|X| \geq Q|\rho| - (Q/2 - 1).$$

If $|\rho| \geq 2$ this exceeds Q . If $\rho = 0$ then $\Delta R = 0$ because $|\Delta R| < Q$, so $k = k'$ and $\delta = 0$; then $|Y| < Q$ forces $e = 0$, hence $s = 0$ and $B = 0$, so the

points coincide. Interchanging the two points if necessary (which negates $\delta, s, \Delta q, \Delta R, A, B$), we may assume

$$(7) \quad \text{rev}_r(k') \equiv \text{rev}_r(k) + 1 \pmod{Q}.$$

The carry pattern. Let $\alpha = \text{rev}_r(k)$ and let t be the number of trailing one-bits of α , with $t = r$ when $\alpha = Q - 1$. For $t < r$, incrementing α clears bits $0, \dots, t-1$ and sets bit t ; in $k = \text{rev}_r(\alpha)$ these are bits $r-1, \dots, r-t$ and $r-1-t$, so

$$(8) \quad \delta = 2^{r-1-t} - (2^r - 2^{r-t}) = -Q + 3 \cdot 2^{r-1-t}.$$

Regular case $t \leq r-3$. The changed bits of k lie at positions ≥ 2 , so $\Delta q = \delta/4$. Put $\beta = 3/2^{t+1}$ and $v = \beta + e - 1$. Then $\delta = Q(\beta - 1)$, and (6) gives $Y = Qv$ and $X \equiv Q(1+v/4) \pmod{N}$. Since $-2 < v \leq 3/2$, the value $Q(1+v/4)$ lies in $[Q/2, 11Q/8] \subset (0, N/2)$ and is the least representative. Hence

$$(9) \quad \frac{X^2 + Y^2}{Q^2} = G(v) := v^2 + \left(1 + \frac{v}{4}\right)^2,$$

a convex quadratic with minimum at $v = -4/17$. The possible values of v are $v = \beta > 0$ if $e = 1$; $v = \beta - 2 \leq -1/2$ if $e = -1$; and, if $e = 0$, $v = 1/2$ for $t = 0$, $v = -1/4$ for $t = 1$, and $v \leq -5/8$ for $t \geq 2$. The corresponding lower bounds are

$$\begin{aligned} G(v) > 1 \quad (e = 1), \quad G(-1/2) &= \frac{65}{64}, \quad G(1/2) = \frac{97}{64}, \\ G(-1/4) &= \frac{241}{256}, \quad G(-5/8) = \frac{1129}{1024}, \end{aligned}$$

using that G decreases on $(-\infty, -4/17]$. Thus $X^2 + Y^2 \geq 241T^2$, with equality exactly when $e = 0$ and $t = 1$.

Carry cases $t \in \{r-2, r-1, r\}$. Here (8) and its wrap-around analogue give $\delta = -Q + c$ with $c = 6, 3, 1$ respectively, and in all three cases $\lfloor k/4 \rfloor = Q/4 - 1$, $\lfloor k'/4 \rfloor = 0$, so $\Delta q = 1 - Q/4$. For $t = r$ the residue is $\Delta R = 1 - Q$, and $Q\Delta R \equiv Q \pmod{N}$ in (6). If $e = 1$, then $X \equiv Q + 1$ is the least representative and $|X| > Q$. If $e = -1$, then $|Y| = 2Q - c > Q$. If $e = 0$, then $X = 3Q/4 + 1$ and $|Y| = Q - c \geq Q - 6$, and

$$(3Q/4 + 1)^2 + (Q - 6)^2 - \frac{241}{256}Q^2 = \frac{159}{256}Q^2 - \frac{21}{2}Q + 37,$$

which equals 28 at $Q = 16$ and increases for $Q \geq 16$. All three cases are strictly longer than $\sqrt{241}T$.

Equality set. Equality requires (7) after orientation, $e = 0$, and $t = 1$, that is, $\text{rev}_r(k) \equiv 1 \pmod{4}$; then $\text{rev}_r(k) + 1 < Q$, so $\Delta R = 1$, and $e = 0$ with $|s| < Q$ gives $s = 0$ and $B = 0$. Conversely every such pair has $X^2 + Y^2 = 241T^2$ by (9), where X is the least representative modulo N ; the coefficient A in (4) is 0 or ± 1 according to whether the first coordinates wrap. There are $Q/4$ admissible values of k and Q values of j , so $Q^2/4 = 2^{m-2}$ unordered pairs attain the minimum. $\square$

Statement	Lean declaration	Scope
Thm. 3.2	<code>net</code>	every even $m \geq 4$, symbolic
Thm. 4.1, upper bound	<code>distance_upper</code>	every even $m \geq 8$, symbolic
Thm. 4.1, value at $m = 8$	<code>distance_m8_exact</code>	all pairs, all $(A, B) \in \mathbb{Z}^2$
Thm. 4.1, count at $m = 8$	<code>attain_m8</code>	64 attaining pairs (kernel)
Thm. 4.1, lower bound, $m \geq 10$	<code>none</code>	paper proof; checks to $m = 30$

TABLE 1. Machine-checked scope (namespace `GKEvenNet`).

Remark 4.2. The hypothesis $m \geq 8$ enters three times: the witness needs $T = Q/16 \in \mathbb{Z}$, $t = 1 \leq r - 3$ requires $r \geq 4$, and the carry-case inequality is evaluated from $Q = 16$. At $m = 4$ and $m = 6$ the minima are 13 and 53, as in Table 1 of [8].

5. VERIFICATION

5.1. Lean formalization. A Lean 4 development [3] over Mathlib [15] (toolchain 4.30.0, Mathlib tag v4.30.0) accompanies the paper. It defines `revr` recursively, the coordinates (1)–(2), with the closed form $Q \text{rev}_r(k)$ of Section 2 for the header’s 32-bit reversal term, and the lattice value $(x' - x + AN + BN/4)^2 + (y' - y + BN)^2$ over $\mathbb{Z}$. Table 1 lists the theorem-bearing declarations. Every declaration depends only on the axioms `propext`, `Classical.choice`, and `Quot.sound`; the finite checks use ordinary kernel reduction (`decide`), not `native_decide`. The proved library `GKEvenNet` contains no `sorry`; `Challenge.lean` states the three theorems with `sorry` as the Palomar template requires, and `Solution.lean` discharges them by importing the library.

The Lean statement of Theorem 3.2 is the classical one: for every even $m = 2r \geq 4$, every $a \leq m$, and every (b_x, b_y) with $b_x < 2^a$ and $b_y < 2^{m-a}$, there is exactly one (k, j) with $k, j < 2^r$ satisfying (3). Its proof follows Section 3 case by case, and a cardinality argument upgrades injectivity to a bijection. The upper bound evaluates the pair of the proof of Theorem 4.1 symbolically in the parameter $(m - 8)/2$. At $m = 8$, a kernel count (`attain_m8`) shows that exactly $64 = 2^{m-2}$ unordered pairs have candidate minimum 241; `attain_m8_spec` states, for every index pair at $m = 8$, that this condition is equivalent to 241 being the minimum of the lattice value over all (A, B) , so exactly 64 pairs attain the tiling minimum, as Theorem 4.1 predicts. The symbolic lower bound for even $m \geq 10$ is left for a revision.

5.2. Independent exact computations. Three programs generate the point set and check the theorems; none imports another. The clean-room Go program transcribes the header literally, including its 32-bit bit reversal, and separately implements (1)–(2); the two agree for every even $m \leq 24$. It checks the net property by counting every dyadic box for even $m \leq 24$, computes all pairwise minima under the ordinary and the row-shifted tiling metric for even $m \leq 16$, checks the minimum $241 \cdot 2^{m-8}$ and the

m	4	6	8	10	12	14	16
row-shifted tiling $\min d_*^2$	13	53	241	964	3856	15424	61696
pairs attaining it	8	16	64	256	1024	4096	16384
ordinary torus $\min d^2$	2	29	185	905	3856	15424	61696
$241 \cdot 2^{m-8}$	–	–	241	964	3856	15424	61696

TABLE 2. All-pairs minima on the integer scale. The first row reproduces the permutation column of Table 1 in [8].

attaining count 2^{m-2} for $m \geq 8$, and checks the witness pair through $m = 30$. A Python reference implementation checks every dyadic box for even $m \leq 16$ and all pairs for even $m \leq 12$. A second Go verifier checks every dyadic box for even $m \leq 24$, all pairs for even $m \leq 12$, and the reduced cases (7) of the distance proof for even $m \leq 30$, checking that exactly $Q/4$ oriented equality cases occur at each m . Each program rejects a point set in which one coordinate has been altered. The $m = 8$ point lists produced by the Lean definitions and by the three programs are identical after newline normalization, with SHA-256 prefix `2ea846f4`; Table 2 lists the computed minima.

5.3. Reproduction. From the root of the reproducibility archive, the script `run_all.sh` in `artifact/` runs the three checkers and the cross-check into `artifact/out/`, leaving the shipped certificates untouched; the artifact README lists the individual commands. The commands are `sh artifact/run_all.sh` and, inside `lean/`, `lake exe cache get && lake build && lake env lean Solution.lean`; on a laptop the Lean build takes about three minutes and the script one.

6. REMARKS

The theorems concern the row-shifted tiling metric. Under the ordinary toroidal metric the computed minima in Table 2 are smaller at $m = 8$ and $m = 10$ and coincide with $241 \cdot 2^{m-8}$ for $m = 12, 14, 16$. Whether the ordinary minimum equals $241 \cdot 2^{m-8}$ for every even $m \geq 12$ is open; the translates with $B = \pm 1$ need $\text{rev}_r(k') - \text{rev}_r(k) \equiv \pm Q/4 + \varepsilon \pmod{Q}$ with $\varepsilon \in \{0, \pm 1\}$, a case outside Section 4.

Table 1 of [8] places the value in context: at $m = 8$ the squared integer-scale minima are 2 for the Hammersley points, 128 for the Larcher–Pillichshammer points [10, 11], 208 for the optimized generator matrices of [7], and 241 for the permutation construction. For N points in a flat torus of area N^2 the hexagonal packing bound gives $d^2 \leq 2N/\sqrt{3} \approx 1.155N$; the even construction attains $d^2 = (241/256)N \approx 0.941N$, and the odd construction attains $d^2 = N$. The even construction replaces the odd construction’s summand k by $\lfloor k/4 \rfloor + S$, its diagonal spacing $2^{\lfloor m/2 \rfloor}$ by S , and the torus by the row-shifted tiling; Lemma 3.1 separates the two bit fields of $x_{k,j}$ that the odd-case argument handles jointly.

Declaration of generative-AI use. Anthropic Claude and OpenAI Codex were used for literature searches, algebraic exploration, proof and checker drafting, the Lean formalization, and adversarial review. The author selected the target and the theorem statements, inspected the source formulas on the rendered page, executed the checks and the Lean build, and accepts full responsibility for the claims; no AI system is an author.

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